October 19, 2026

1 Recall: sections, rational functions, and projective maps

Work over $\mathbb{C}$. Sections of a line bundle $L \to X$ are geometric objects: a section chooses one vector in each fiber. They become easier to calculate with after choosing a nonzero rational section s_D. If $\operatorname{div}(s_D) = D$, then every rational section has a unique expression

$$\widetilde{s} = f s_D, \qquad f \in \mathbb{C}(X),$$

and it is regular precisely when

$$\operatorname{div}(f) + D \geq 0.$$

Thus a regular section can be represented by a rational function with poles: the zeros of the chosen reference section compensate for them.

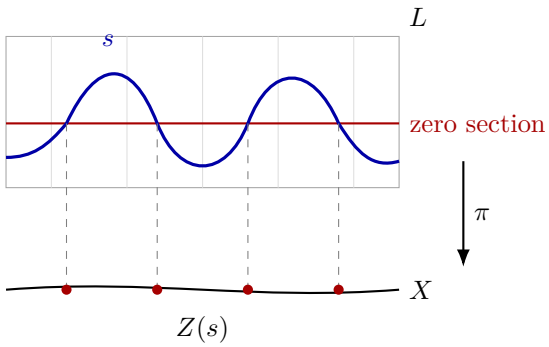


Diagram 1: A schematic regular section s. Its intersections with the zero section project to its zero divisor $Z(s)$ on X. The picture illustrates the regular case; a rational reference section may also have poles.

1.1 Three descriptions of the same projective coordinates

Suppose $s_0, \ldots, s_r$ are global sections of L. At a point x where they do not all vanish, their values lie in the same one-dimensional vector space $L|_x$, so they determine

$$\varphi(x) = [s_0(x) : \cdots : s_r(x)] \in \mathbb{P}^r.$$

Changing the identification $L|_x \cong \mathbb{C}$ rescales every coordinate by the same scalar. In terms of rational functions,

$$s_i = f_i s_D \quad \Longrightarrow \quad \varphi = [f_0 : \cdots : f_r]$$

where this rational-function formula can be evaluated. On the open set $s_j \neq 0$, the regular affine coordinates are instead

$$\left(\frac{s_0}{s_j}, \ldots, \widehat{\frac{s_j}{s_j}}, \ldots, \frac{s_r}{s_j} \right).$$

The hat means that the coordinate equal to 1 is omitted. These are equivalent descriptions, provided one keeps track of the chosen frame and of where the denominator does not vanish. A line bundle is basepoint free when its full space of global sections has no common zero.

2 Dividing by a section translates the polytope

For a torus-invariant Cartier divisor D on X_Σ, write

$$L = \mathcal{O}_{X_\Sigma}(D), \qquad P_D = \{m \in M_{\mathbb{R}} : \langle m, u_\rho \rangle \geq -a_\rho \text{ for all } \rho\}\,.$$

The global monomial sections are

$$s_m = \chi^m s_D, \qquad m \in P_D \cap M.$$

The lattice points describe sections regular *everywhere* on X_Σ, not merely on a single affine chart.

We return to the trapezoid from October 16:

$$P = \operatorname{conv}\{(0,0),(2,0),(1,1),(0,1)\} = \{(r,q) : r,q \geq 0,\ q \leq 1,\ r+q \leq 2\}\,.$$

Its five lattice points give

$$\varphi_P : (\mathbb{C}^\times)^2 \longrightarrow \mathbb{P}^4, \qquad (s,t) \longmapsto [1 : s : t : s^2 : st].$$

In this example the map extends to an embedding of the normal-fan variety, as the chart calculations from October 16 showed. The point of today's discussion is why that conclusion requires a check.

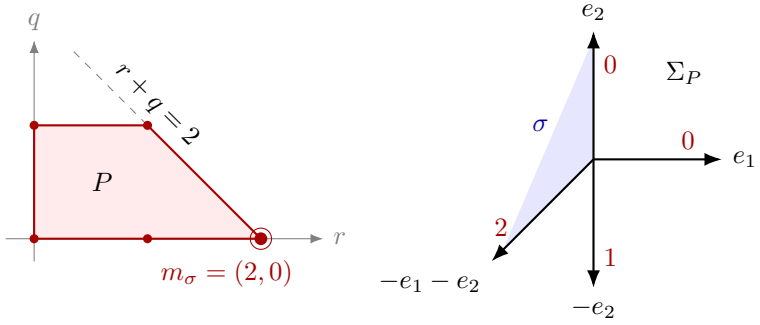


Diagram 2: The trapezoid P and its inward normal fan. The red numbers on the rays are the divisor coefficients a_ρ. The shaded cone $\sigma = \operatorname{Cone}(e_2, -e_1 - e_2)$ corresponds to the marked vertex $m_\sigma = (2,0)$.

2.1 The local frame on an affine chart

For a maximal cone σ, the Cartier datum m_σ satisfies

$$\langle m_\sigma, u_\rho \rangle = -a_\rho \qquad (\rho \subseteq \sigma).$$

The section

$$e_\sigma = \chi^{m_\sigma} s_D$$

is a nowhere-vanishing frame on the affine chart X_σ. Consequently,

$$H^0(X_\sigma, L) \xrightarrow{\widetilde{s} \mapsto \widetilde{s}/e_\sigma} \mathcal{O}(X_\sigma)$$

is an isomorphism of $\mathcal{O}(X_\sigma)$-modules. For a monomial section this becomes

$$\boxed{\frac{s_m}{e_\sigma} = \chi^{m - m_\sigma}.}$$

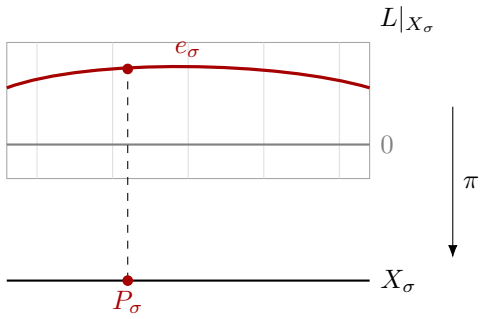


Diagram 3: The local frame e_σ never meets the zero section. It identifies $L|_{X_\sigma}$ with $X_\sigma \times \mathbb{A}^1$; dividing by it turns a regular section into a regular function.

Thus dividing sections by the distinguished local frame means subtracting m_σ from every exponent.

In the trapezoid example, $m_\sigma = (2, 0)$, so $e_\sigma = s^2 s_D$. Dividing the five global sections by this frame gives

$$(s^{-2}, s^{-1}, t/s^2, 1, t/s).$$

The section indexed by m_σ becomes the constant 1. This is the section that does not vanish at the fixed point P_σ.

2.2 The dual cone records all local regular monomials

Translation puts the vertex at the origin, and

$$P - m_\sigma \subseteq \sigma^\vee = \mathrm{Cone}((-1, 0), (-1, 1)) = \{(a, b) : b \geq 0,\ a + b \leq 0\}.$$

The affine chart is

$$X_\sigma = \operatorname{Spec} \mathbb{C}[\sigma^\vee \cap M],$$

so the lattice points of the *whole cone* index its regular monomials. In this case, every such lattice point has the expression

$$(a, b) = (-a - b)(-1, 0) + b(-1, 1).$$

The two generators form a lattice basis, and hence

$$\mathbb{C}[\sigma^\vee \cap M] = \mathbb{C}[s^{-1}, t/s].$$

With $\xi = s^{-1}$ and $\eta = t/s$, the normalized affine tuple is

$$(\xi^2, \xi, \xi\eta, \eta),$$

which identifies the image chart with $X_\sigma \cong \mathbb{A}^2$.

This answers the question about the dots. Points of $P - m_\sigma$ come from global sections after dividing by the frame. The extra cone points give local functions, but multiplying them by the frame need not produce sections regular on every other chart. For example, s^{-3} is regular on X_σ, whereas

$$s^{-3} e_\sigma = s^{-1} s_D$$

is not a global section, since $(-1, 0) \notin P$. The variety is denoted X_σ; $\sigma^\vee$ describes its coordinate ring, not a different chart $X_{\sigma^\vee}$.

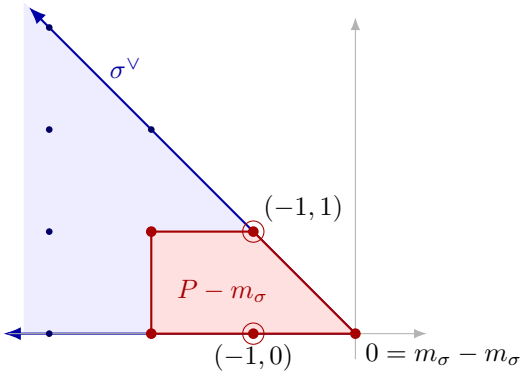


Diagram 4: Dividing monomials by χ^{m_σ} translates P into $\sigma^\vee$. Red points come from global monomial sections. All cone lattice points give local regular functions; the two circled nonzero points generate their semigroup.

3 Ample does not always mean very ample

A full-dimensional lattice polytope $P \subseteq M_{\mathbb{R}}$ determines an inward normal fan Σ_P, a normal projective toric variety X_{Σ_P}, and an ample Cartier divisor D_P. Its lattice points give a basepoint-free complete linear system, but the map

$$\varphi_{D_P} : X_{\Sigma_P} \longrightarrow \mathbb{P}^{|P \cap M|-1}$$

need not be a closed embedding.

The chart calculation explains the additional requirement. At a vertex v, the normalized sections have exponents $m - v$. To recover the entire affine chart, they must generate its semigroup:

$$\mathbb{N}\{m - v : m \in P \cap M\} = \mathrm{Cone}(P - v) \cap M.$$

Here $\mathbb{N}\{\cdots\}$ means all nonnegative integral combinations. It is not enough that the vectors generate the same real cone. Checking this equality at every vertex is the vertex-chart criterion for very ampleness.

In dimension two every lattice polygon satisfies this condition (indeed, every lattice polygon is normal). Thus its associated divisor is very ample. The next example shows what can go wrong in dimension three. Smoothness of the variety is a separate issue from whether a given linear system embeds it.

4 A three-dimensional example

Put $M = \mathbb{Z}^3$, let e_1, e_2, e_3 be its standard basis, and set

$$w = (1, 1, 2), \qquad P = \mathrm{conv}\{0, e_1, e_2, w\}.$$

This is the height-two Reeve tetrahedron. Its only lattice points are its four vertices.

The lattice-point map is

$$\varphi_P : (\mathbb{C}^\times)^3 \longrightarrow \mathbb{P}^3, \qquad (s, t, u) \longmapsto [1 : s : t : stu^2].$$

Since

$$\varphi_P(s, t, u) = \varphi_P(s, t, -u),$$

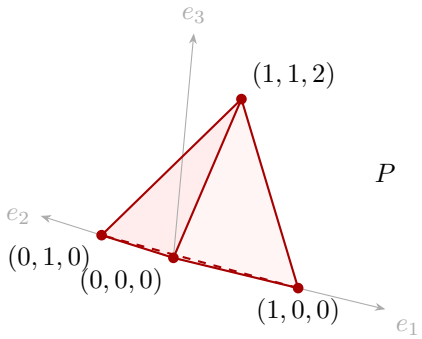


Diagram 5: The tetrahedron $P = \mathrm{conv}\{(0,0,0),(1,0,0),(0,1,0),(1,1,2)\}$. Its four vertices are its only lattice points. The dashed edge is hidden by the two shaded faces.

it is not injective. In fact its image is the entire dense torus of $\mathbb{P}^3$: for any nonzero a, b, c, choose u with $u^2 = c/(ab)$. Therefore

$$\boxed{\overline{\varphi_P(T)} = \mathbb{P}^3.}$$

The equality that fails is the identification of this image with X_{Σ_P} for the prescribed lattice $M = \mathbb{Z}^3$. The exponent differences generate only

$$M' = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}w = \left\{(a,b,c) \in \mathbb{Z}^3 : c \text{ is even}\right\},$$

an index-two sublattice of M. The induced morphism $X_{\Sigma_P} \to \mathbb{P}^3$ is finite of degree two. It is not an embedding, and it is not a normalization map: $\mathbb{P}^3$ is already normal.

4.1 The missing local monomial and the singular chart

At the vertex 0, the lattice point

$$h = (1,1,1) = \tfrac{1}{2}(e_1 + e_2 + w)$$

lies in $\mathrm{Cone}(P)$, but not in the semigroup generated by $P \cap M$. The four vectors e_1, e_2, w, h form the Hilbert basis of $\mathrm{Cone}(P) \cap M$, meaning its minimal set of additive generators. The missing character is $\chi^h = stu$.

Consequently the chart of X_{Σ_P} at this vertex is

$$\mathrm{Spec}\,\mathbb{C}[s, t, stu^2, stu] \cong \mathrm{Spec}\,\mathbb{C}[a,b,c,d]/(d^2 - abc).$$

Its origin is singular: the hypersurface has dimension three, and all first derivatives of $d^2 - abc$ vanish there. In contrast, the chart seen by the original four coordinates has ring $\mathbb{C}[s, t, stu^2] \cong \mathbb{C}[a,b,c]$, an affine three-space. Thus the normal-fan variety can be singular even though this particular image is the smooth variety $\mathbb{P}^3$.

4.2 Doubling the polytope supplies the missing generators

The dilation $2P$ has the same normal fan as P and corresponds to $2D_P$. Its eleven lattice points and their monomials are listed below; the order keeps the board's coordinates X_3, X_4, X_5.

coordinate	lattice point	monomial
X_0	$(0,0,0)$	1
X_1	$(1,0,0)$	s
X_2	$(0,1,0)$	t
X_3	$(1,1,0)$	st
X_4	$(1,1,1)$	stu
X_5	$(1,1,2)$	stu^2
X_6	$(2,0,0)$	s^2
X_7	$(0,2,0)$	t^2
X_8	$(2,1,2)$	s^2tu^2
X_9	$(1,2,2)$	st^2u^2
X_{10}	$(2,2,4)$	$s^2t^2u^4$

The two points $(2,1,2)$ and $(1,2,2)$, omitted from the quick board list, are edge midpoints and must be included in the complete linear system.

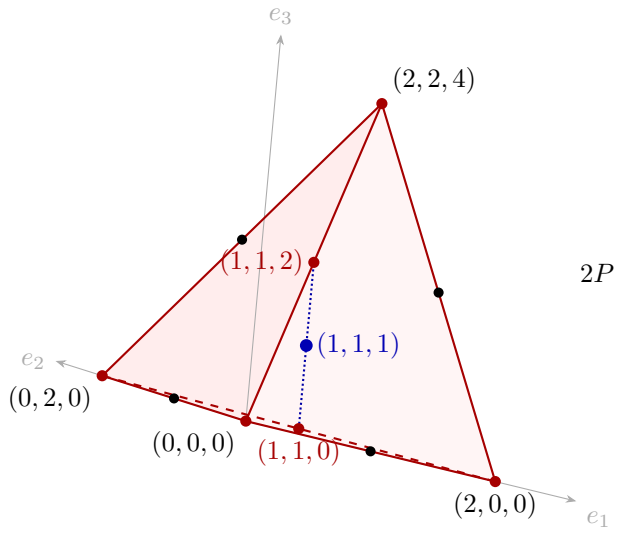


Diagram 6: All eleven lattice points of $2P$, including the two edge midpoints $(2,1,2)$ and $(1,2,2)$. The highlighted collinear points satisfy $2(1,1,1) = (1,1,0) + (1,1,2)$, yielding the homogeneous relation $X_4^2 = X_3X_5$ for the three corresponding coordinates.

The complete map now targets $\mathbb{P}^{10}$. On the torus, the new coordinate recovers the missing sign:

$$u = \frac{stu}{st} = \frac{X_4}{X_3}.$$

The three highlighted collinear lattice points also give the homogeneous equation

$$\boxed{X_4^2 = X_3X_5.}$$

This is one relation on the image, not a complete set of its equations.

To prove an embedding, we must also check the boundary charts. Let v be any vertex of P, and write

$$r_i = v_i - v \quad (i = 1, 2, 3)$$

for its differences to the other three vertices. Their determinant has absolute value two, and

$$h_v = h - 2v = \tfrac{1}{2}(r_1 + r_2 + r_3) \in M.$$

The half-open fundamental parallelepiped of the r_i contains exactly the two lattice points $0, h_v$. Subtracting integral parts of the cone coefficients therefore gives

$$\mathrm{Cone}(P - v) \cap M = \mathbb{N}r_1 + \mathbb{N}r_2 + \mathbb{N}r_3 + \mathbb{N}h_v.$$

Each generator occurs among the normalized sections at the vertex $2v$ of $2P$:

$$r_i = (v + v_i) - 2v, \qquad h_v = h - 2v, \qquad v + v_i,\ h \in 2P \cap M.$$

Thus the complete sections of $2D_P$ generate every affine chart coordinate ring. The vertex-chart criterion proves that

$$X_{\Sigma_P} \hookrightarrow \mathbb{P}^{10}$$

is a closed embedding. Hence D_P is ample but not very ample, whereas $2D_P$ is very ample. Doubling repairs the linear system; it leaves the normal fan and the singularities of the variety unchanged.

5 Ampleness, normal fans, and projectivity

Let X_Σ be a normal complete toric variety and D a torus-invariant Cartier divisor. Recall that D is ample when nD is very ample for some integer $n > 0$. The fan formulation is

$$\boxed{D \text{ ample} \quad \Longleftrightarrow \quad P_D \text{ full dimensional and } \Sigma_{P_D} = \Sigma.}$$

Thus the polytope must recover the original fan, with all its cones and walls.

For a basepoint-free divisor with full-dimensional P_D, the original fan Σ refines the normal fan Σ_{P_D}: each cone of Σ lies in a cone of Σ_{P_D}. The latter can be a *coarser* fan, with fewer walls. Neighboring cones can merge, and a ray of Σ can lie in the interior of a higher-dimensional cone of Σ_{P_D}. The basepoint-free hypothesis matters here. If P_D has smaller dimension, its normal fan is considered in the quotient dual to $\mathrm{span}_{\mathbb{R}}(P_D - P_D)$.

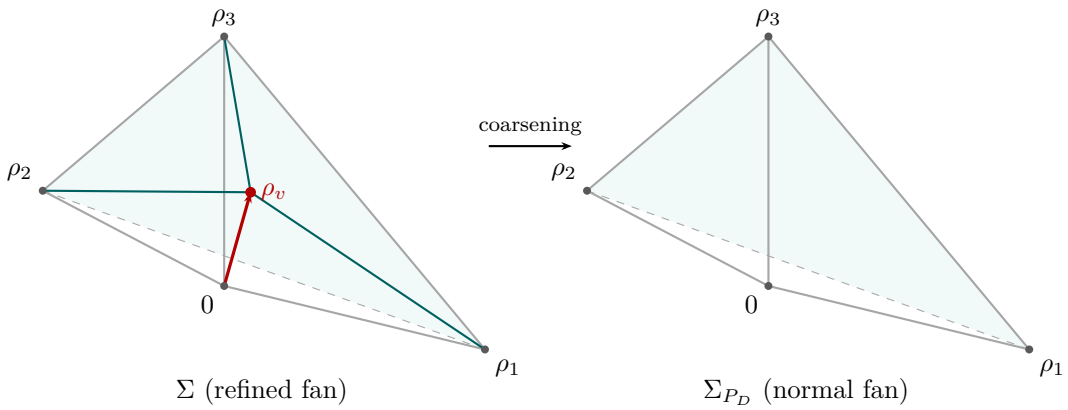


Diagram 7: A triangular slice of one three-dimensional cone of the normal fan of a simplex. Adding the interior ray $\rho_v = \mathbb{R}_{\geq 0}(u_1 + u_2 + u_3)$ subdivides this cone into three cones; the other cones of the complete fan are unchanged. The identity lattice map induces the coarsening morphism $X_\Sigma \longrightarrow X_{\Sigma_{P_D}}$. This is a schematic example, separate from the preceding tetrahedron.

5.1 Strict convexity remembers every wall

Use the classroom sign convention from October 16:

$$q_\sigma = -m_\sigma, \qquad \psi_D(u) = \langle q_\sigma, u\rangle \quad (u \in \sigma), \qquad \psi_D(u_\rho) = a_\rho.$$

Basepoint freeness is equivalent to the maximum formula

$$\psi_D(u) = \max_{\sigma \in \Sigma(\dim X)} \langle q_\sigma, u\rangle \,.$$

Assume this convexity condition holds. If adjacent cones have the same linear piece, the function does not notice their common wall. Strict convexity requires a genuine bend at every wall: its maximal domains of linearity must be exactly the maximal cones of Σ. Equivalently, for every maximal cone σ,

$$\psi_D(u) = \langle q_\sigma, u\rangle \quad \Longleftrightarrow \quad u \in \sigma.$$

Here strict convexity is relative to the fan; the function is still linear on each cone. We obtain

$$D \text{ ample} \quad \Longleftrightarrow \quad \psi_D \text{ strictly convex.}$$

With the opposite support-function convention, one negates ψ_D and uses a minimum instead; the geometric criterion is unchanged.

Remark 1 (The question about the canonical divisor). Being $\mathbb{Q}$-Gorenstein means that the canonical divisor K_X is $\mathbb{Q}$-Cartier: some positive integer multiple of K_X is Cartier. This is a condition on the canonical divisor, distinct from ampleness of a chosen divisor D. The latter asks whether a positive multiple of D gives a projective embedding.

5.2 How to ask whether a toric variety is projective

First ask for completeness: a projective variety is complete. For a toric variety this means

$$|\Sigma| = N_{\mathbb{R}}.$$

Over $\mathbb{C}$, this is equivalent to compactness in the complex analytic topology. Next ask whether the fan admits an *integral strictly convex support function*. Such a function produces an ample Cartier divisor. Hence

$$X_\Sigma \text{ projective} \iff \begin{array}{c}\Sigma \text{ is complete and admits an} \\ \text{integral strictly convex support function.}\end{array}$$

Completeness alone is insufficient in dimensions three and higher. For surfaces, however, the second condition follows automatically.

6 The surface dictionary

The following discussion concerns normal toric surfaces. For the calculations, return to the two-dimensional trapezoid from Diagram 2:

$$P = \operatorname{conv}\{(0,0),(2,0),(1,1),(0,1)\}.$$

Write D for its associated very ample divisor on X_{Σ_P}.

1. **Completeness is equivalent to projectivity.** Every complete two-dimensional fan is the normal fan of a lattice polygon. Thus every normal complete toric surface is projective.

   The classroom picture is *pitching a tent*: choose pole heights on the rays so that a piecewise-linear bedsheet bends strictly across every wall. This asserts that suitable heights *exist*; arbitrary heights need not be convex.

   One construction explains the existence. List the primitive rays $u_i = (a_i, b_i)$ in counterclockwise order. Completeness gives positive rational numbers λ_i with $\sum_i \lambda_i u_i = 0$: express each $-u_i$ as a nonnegative rational combination of rays and sum the resulting relations. Clear denominators. The integral edge vectors

   $$\lambda_i(b_i, -a_i)$$

   then form a closed convex polygonal chain. Its inward normals are precisely the u_i, so the resulting lattice polygon has the prescribed normal fan.

2. **Smoothness is visible at the vertices.** At a vertex p, take the two *primitive* lattice vectors $u, v \in M$ pointing from p along its adjacent edges. The corresponding affine chart is smooth exactly when

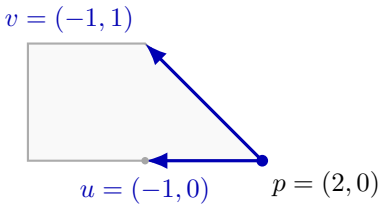
   $$|\det(u, v)| = 1.$$

   In other words, these directions must form a basis of M. The surface is smooth when this holds at every vertex. These are directions in the tangent cone in $M_\mathbb{R}$; its dual is the corresponding normal fan cone in $N_\mathbb{R}$.

   At the marked vertex, $u = (-1, 0)$ and $v = (-1, 1)$ give determinant -1. The other three vertices also pass the primitive-vector test, so this surface is smooth. Using the full bottom-edge vector $(-2, 0)$ would give the wrong smoothness test.

3. **Intersection with an invariant curve is an edge length.** For a ray ρ, let F_ρ be the edge of P_D with primitive inward normal u_ρ. The corresponding torus-invariant divisor D_ρ is a curve, and

   $$\boxed{D \cdot D_\rho = \operatorname{Length}_M(F_\rho).}$$



$$\det \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix} = -1$$

Diagram 8: The primitive edge directions at $p = (2, 0)$ form a lattice basis. The full bottom edge has length two, but its primitive direction is $(-1, 0)$.

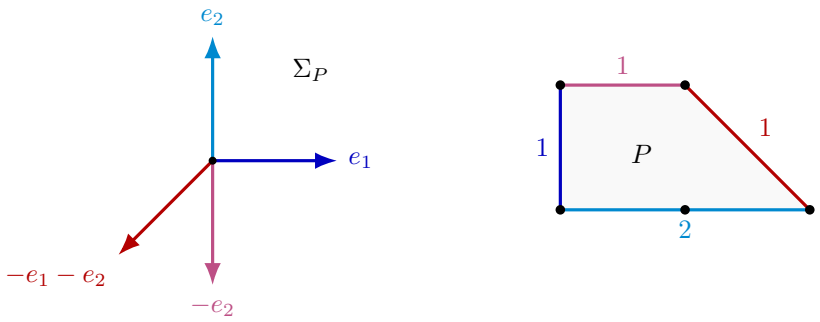


Diagram 9: Matching colors identify a fan ray with the polygon edge whose inward normal it gives. The numbers on the edges are their lattice lengths, hence the intersections $D \cdot D_\rho$.

For endpoints p, q, write $q - p = \ell w$ with w primitive; then the lattice length is ℓ, or equivalently $|F_\rho \cap M| - 1$. The length is measured in the lattice, not by Euclidean distance.

The four intersection numbers are

u_ρ	F_ρ	$D \cdot D_\rho$
e_1	$r = 0$	1
e_2	$q = 0$	2
$-e_2$	$q = 1$	1
$-e_1 - e_2$	$r + q = 2$	1

Thus dark blue gives 1, light blue gives 2, pink gives 1, and red gives 1, as on the board.

4. **The degree is twice the lattice area.** If a very ample divisor D embeds the surface by φ_D, two general hyperplanes pull back to divisors in $|D|$. Their intersection number is the degree of the image:

$$\deg \varphi_D(X_\Sigma) = D^2 = 2 \operatorname{Area}_M(P_D).$$

Here a fundamental lattice parallelogram has area 1. The factor $2 = 2!$ converts that area into normalized lattice volume.

One way to see the formula is to compare leading coefficients: lattice-point counting gives

$$h^0(X_\Sigma, \mathcal{O}(kD)) = |kP_D \cap M| = \operatorname{Area}_M(P_D)k^2 + O(k),$$

while the Hilbert polynomial has leading term $\frac{1}{2}D^2k^2$. For our trapezoid,

$$\operatorname{Area}_M(P) = \frac{2+1}{2} \cdot 1 = \frac{3}{2}, \qquad \boxed{D^2 = 3.}$$

Therefore the surface embedded in $\mathbb{P}^4$ by the five monomial sections has degree 3.

7 The Hirzebruch surface behind Diagram 9

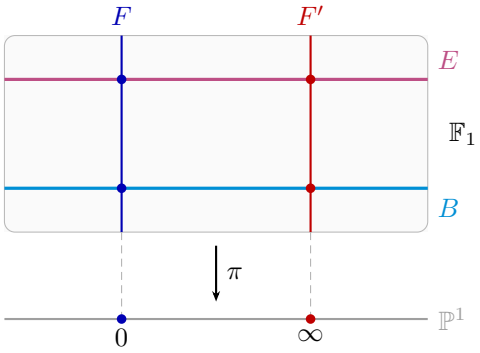
The surface in Diagram 9 is the first Hirzebruch surface

$$X = X_{\Sigma_P} \cong \mathbb{F}_1 \cong \mathrm{Bl}_p \mathbb{P}^2.$$

The following names identify the four boundary curves by their colors and primitive ray generators:

curve	u_ρ	geometric role
F	$(1,0)$	fiber
B	$(0,1)$	positive section
F'	$(-1,-1)$	fiber
E	$(0,-1)$	negative section

There is a ruling $\pi : \mathbb{F}_1 \to \mathbb{P}^1$, given on the torus by $(s,t) \mapsto s$. Its two invariant fibers are F, F', over $0, \infty$, respectively. The curves E, B are disjoint sections, each meeting every fiber once. The curve E is also the exceptional curve of the blowup.



Continuing Diagram 9. Schematic of the ruling $\pi : \mathbb{F}_1 \to \mathbb{P}^1$. The sections E and B are disjoint; the fibers $F = \pi^{-1}(0)$ and $F' = \pi^{-1}(\infty)$ each meet each section once.

7.1 The class group relations come from characters

For a toric character, the principal divisor formula is

$$\mathrm{div}(\chi^m) = \sum_\rho \langle m, u_\rho \rangle \, D_\rho.$$

Apply it to $s = \chi^{(1,0)}$ and $t = \chi^{(0,1)}$:

$$\mathrm{div}(s) = F - F', \qquad \mathrm{div}(t) = B - F' - E.$$

Principal divisors represent zero in the divisor class group. Consequently,

$$F' \sim F, \qquad B \sim E + F.$$

The board's formula $B = E + F$ is this equality of divisor classes; $\sim$ denotes linear equivalence.

The invariant divisors generate $\mathrm{Cl}(X)$, and the character divisors give all their relations. Explicitly,

$$\mathrm{Cl}(X) = \frac{\mathbb{Z}F \oplus \mathbb{Z}B \oplus \mathbb{Z}F' \oplus \mathbb{Z}E}{\langle F - F',\ B - F' - E \rangle} \cong \mathbb{Z}[F] \oplus \mathbb{Z}[E].$$

Eliminating F' and B leaves the two free generators $[F], [E]$. Since X is smooth, every Weil divisor is Cartier, so

$$\boxed{\mathrm{Pic}(\mathbb{F}_1) = \mathrm{Cl}(\mathbb{F}_1) \cong \mathbb{Z}[F] \oplus \mathbb{Z}[E].}$$

Intersection numbers depend only on these classes and are bilinear.

7.2 Recovering the intersection table

Distinct fibers are disjoint, and $F' \sim F$, so

$$F^2 = F \cdot F' = 0.$$

Each section meets a fiber transversely once:

$$E \cdot F = 1, \qquad B \cdot F = 1.$$

The sections E, B are disjoint. Substituting $B \sim E + F$ therefore gives

$$0 = E \cdot B = E \cdot (E + F) = E^2 + 1, \qquad E^2 = -1.$$

Finally,

$$B^2 = (E + F)^2 = -1 + 2 + 0 = 1.$$

Thus the board's list is

$$\boxed{F^2 = 0, \quad E^2 = -1, \quad B^2 = 1, \quad E \cdot B = 0, \quad E \cdot F = B \cdot F = 1.}$$

In the ordered basis $([F], [E])$, the intersection matrix is

$$\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}.$$

The schematic picture supplies the intersections of distinct curves; linear equivalence then determines the self-intersections.

7.3 Recovering the four edge lengths in Diagram 9

The divisor coefficients on the fan are $0, 0, 2, 1$, in the order (F, B, F', E). Thus the polygon's actual invariant divisor is

$$D = 2F' + E, \qquad D \sim 2F + E.$$

Using the intersection table, we recover every colored edge length:

$$\begin{aligned} D \cdot F &= (2F + E) \cdot F = 1 && \text{(dark blue)}, \\ D \cdot B &= (2F + E) \cdot B = 2 && \text{(light blue)}, \\ D \cdot E &= (2F + E) \cdot E = 2 - 1 = 1 && \text{(pink)}, \\ D \cdot F' &= D \cdot F = 1 && \text{(red)}. \end{aligned}$$

In particular, $D \cdot E = 1$ and $E^2 = -1$ are different products. The first is the degree of the chosen line bundle along E; the second is the degree of $\mathcal{O}_X(E)|_E$.

The self-intersection of the embedding divisor is

$$D^2 = (2F + E)^2 = 4F^2 + 4F \cdot E + E^2 = 3,$$

agreeing with twice the area of the trapezoid. If H is the pullback of the line class under $\mathrm{Bl}_p \mathbb{P}^2 \to \mathbb{P}^2$, then $B \sim H$ and $F \sim H - E$. Consequently

$$D \sim 2F + E \sim 2H - E,$$

connecting this example with the earlier five-section embedding of the blowup into $\mathbb{P}^4$.

Remark 2 (The other Hirzebruch surfaces). For $r \geq 0$, replacing the red ray by $(-1, -r)$ gives the fan of $\mathbb{F}_r$. If E denotes the section of self-intersection $-r$, and F a fiber, then

$$\mathrm{Pic}(\mathbb{F}_r) = \mathrm{Cl}(\mathbb{F}_r) = \mathbb{Z}[E] \oplus \mathbb{Z}[F], \qquad B \sim E + rF,$$

$$F^2 = 0, \qquad E \cdot F = 1, \qquad E^2 = -r, \qquad B^2 = r, \qquad E \cdot B = 0.$$

Hence a class $aE + bF$ has square $2ab - ra^2$. The Hirzebruch index r records the self-intersection of E; the degree of a projective embedding also depends on the chosen divisor. Our surface is $\mathbb{F}_1$, with embedding degree 3.

8 Intersections by restricting the line bundle

Let X be a smooth projective surface, $C \subset X$ a smooth complete curve, and $L = \mathcal{O}_X(D)$. The line bundle always restricts to $L|_C$, and

$$\boxed{D \cdot C = \deg(L|_C).}$$

When L has a nonzero regular section, choose one, $\widetilde{s}$. Its zero divisor $Z(\widetilde{s})$ is linearly equivalent to D. Its zero set is where the graph of the section meets the zero section of the bundle.

If C is not a component of $Z(\widetilde{s})$, then $\widetilde{s}|_C \not\equiv 0$, and its zeros are the intersection with C. Thus

$$D \cdot C = \sum_{p \in C \cap Z(\widetilde{s})} \mathrm{ord}_p(\widetilde{s}|_C).$$

An intersection transverse to C contributes 1; otherwise its multiplicity must be counted.

If C is a component of $Z(\widetilde{s})$, the restricted section is identically zero. The line bundle $L|_C$ still exists, but this section does not give a finite divisor on C. When a regular replacement is available, choose one whose restriction is nonzero. A basepoint-free system guarantees such a choice; over $\mathbb{C}$, a general section has simple zeros on C. In our example D is very ample, so this choice is available.

8.1 Changing the section adds a principal divisor

Any two nonzero rational sections of the same line bundle differ by a nonzero rational function:

$$\widetilde{s}' = g\widetilde{s}, \qquad \mathrm{div}(\widetilde{s}') = \mathrm{div}(\widetilde{s}) + \mathrm{div}(g).$$

This is the meaning of moving a divisor within its linear equivalence class. The line bundle remains the same.

For an arbitrary line bundle, a regular replacement avoiding C need not exist. One can instead choose a rational section with neither a zero nor a pole along the generic point of C. Its restriction is a nonzero rational section of $L|_C$, and the degree is the number of zeros *minus* the number of poles, with multiplicities. Multiplication by a rational function produces another regular global section precisely when its resulting section divisor is effective. The toric choices below satisfy this condition.

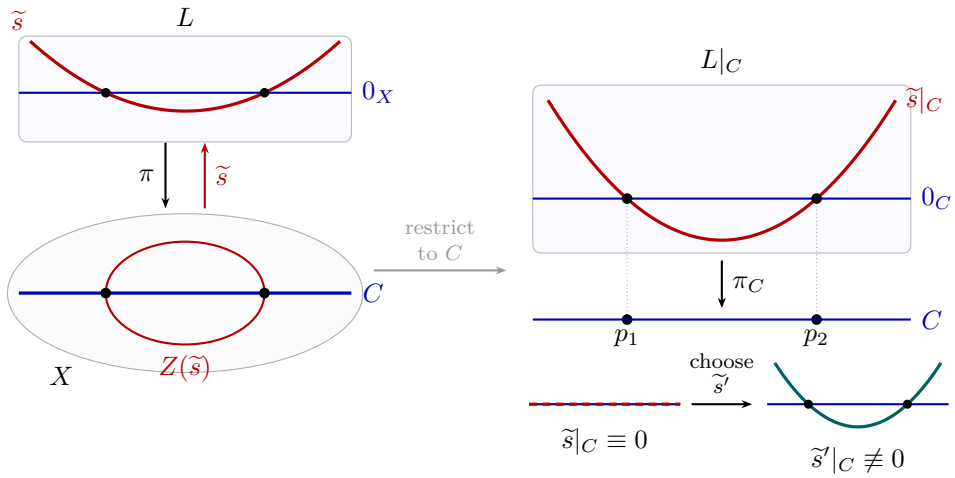


Diagram 10: Restricting a line bundle and a section to a smooth projective curve. When $\widetilde{s}|_C \not\equiv 0$, its zero divisor is finite and is obtained by restricting $Z(\widetilde{s})$ to C, with multiplicities. The lower sketches contrast a section vanishing identically on C with a replacement section whose restriction is nonzero. The bundle panels are schematic slices; the crossings do not assert transversality.

9 Restricting to a torus-invariant curve

Continue with the fan and curve names from Diagram 9:

$$F = D_{(1,0)}, \quad B = D_{(0,1)}, \quad F' = D_{(-1,-1)}, \quad E = D_{(0,-1)}, \qquad D = 2F' + E.$$

For a ray ρ, the one-dimensional orbit is $O(\rho) \cong \mathbb{C}^\times$. Its *closure*

$$C = V(\rho) = D_\rho \cong \mathbb{P}^1$$

is the complete invariant curve used in the intersection calculation.

The fan of C is obtained in two steps: keep the cones containing ρ, then pass to the quotient lattice

$$N_\rho = N/\mathbb{Z}u_\rho.$$

Here u_ρ is primitive. The images of those cones form the star fan

$$\Sigma_C = \{\overline{\sigma} : \rho \subseteq \sigma \in \Sigma\}.$$

In this surface example it is the complete one-dimensional fan with two rays. Curves corresponding to rays outside this star need not contribute, even if their vectors have nonzero images in the quotient.

9.1 The coefficient on the chosen curve is already zero

First take $C = F$, the dark-blue curve. Its coefficient in $D = 2F' + E$ is 0, so the canonical section s_D does not vanish identically on F. The two maximal cones containing its ray are

$$\mathrm{Cone}(e_1, e_2), \qquad \mathrm{Cone}(-e_2, e_1).$$

The quotient map is $q(a, b) = b$. Its neighboring rays give

$$e_2 \longmapsto 1, \qquad -e_2 \longmapsto -1.$$

Their divisor coefficients are 0 and 1. Therefore

$$D|_F = [E \cap F], \qquad L|_F \cong \mathcal{O}_{\mathbb{P}^1}(1), \qquad D \cdot F = 1.$$

The red curve F' is disjoint from F. Although its vector $(-1,-1)$ also maps to -1, its cones are not in this star: projecting all rays would incorrectly include it.

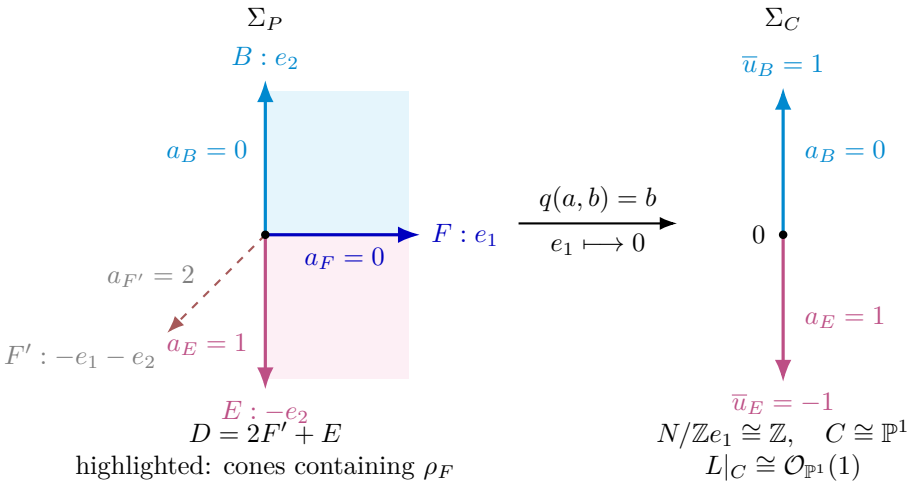


Diagram 11: Restricting $L = \mathcal{O}(D)$ to the dark-blue invariant curve $C = F = V(\rho_F)$. Its fan is obtained by taking the cones containing ρ_F and passing to $N/\mathbb{Z}e_1$. Since $a_F = 0$, the neighboring coefficients $a_B = 0$ and $a_E = 1$ descend directly: the restricted divisor is one point on $\mathbb{P}^1$, so $\deg L|_C = 1$. The curve corresponding to the greyed red ray is disjoint from C and contributes nothing.

9.2 Removing a component before restricting

Now take $C = F'$, whose coefficient in D is 2. The original section s_D vanishes identically on C. Use either of the following sections of the same line bundle:

$$\widetilde{s}_1 = st\, s_D, \qquad \widetilde{s}_2 = s^2 s_D.$$

The principal-divisor relations from the previous section give

$$A_1 := D + \operatorname{div}(st) = F + B,$$
$$A_2 := D + \operatorname{div}(s^2) = 2F + E.$$

Both divisors are effective, and neither contains F'. Their coefficients, in the original fan order, are

	F	B	F'	E
D	0	0	2	1
A_1	1	1	0	0
A_2	2	0	0	1

This is what the board's linear adjustments mean. In the classroom convention $\psi_D(u_\rho) = a_\rho$,

$$\psi_{D+\operatorname{div}(\chi^m)}(u) = \psi_D(u) + \langle m, u \rangle .$$

Writing $u = (x, y)$, multiplication by st adds $x + y$; multiplication by s^2 adds $2x$. Both linear forms have value -2 at $(-1, -1)$, so the new coefficient there is 0. The relevant linear pieces now annihilate that ray and descend to the quotient *on its star*.

For $C = F'$, the quotient map is $q'(a, b) = b - a$. Again B maps to $+1$ and E to -1; the disjoint fiber F contributes nothing. Set

$$p_B = B \cap F', \qquad p_E = E \cap F'.$$

Then the two restrictions are

$$A_1|_{F'} = p_B, \qquad A_2|_{F'} = p_E.$$

Their coefficient pairs on the quotient fan are $(1, 0)$ and $(0, 1)$, respectively. Each is a point divisor on $\mathbb{P}^1$, so both represent the same line bundle of degree 1. Indeed, the quotient character $z = (t/s)|_{F'}$ satisfies

$$\mathrm{div}_{F'}(z) = p_B - p_E.$$

Thus the two choices differ by a principal divisor on the curve, just as the two sections differ by t/s.

9.3 The restricted polytope is the corresponding edge

The edge of P_D dual to F' is

$$G = \{(r, q) \in P_D : r + q = 2\} = [(2, 0), (1, 1)].$$

The two endpoints index the sections $s^2 s_D$ and $st\, s_D$; their restrictions to F' are not identically zero.

Changing the divisor by $\mathrm{div}(\chi^m)$ translates its polytope by $-m$. The lattice of characters on F' is

$$M \cap (-1, -1)^\perp = \mathbb{Z}(-1, 1).$$

In this rank-one lattice, the two choices above give

$$G - (1, 1) = [(1, -1), (0, 0)] \cong [-1, 0],$$
$$G - (2, 0) = [(0, 0), (-1, 1)] \cong [0, 1].$$

These intervals differ by a lattice translation and both have length 1. This is the polytope version of the same calculation:

$$\boxed{D \cdot F' = \deg(L|_{F'}) = \mathrm{Length}_M(G) = 1.}$$

Remark 3 (A negative degree requires poles). The distinction between regular and rational sections also explains $E^2 = -1$. For the line bundle $\mathcal{O}_X(E)$, the original section vanishes on E. Since

$$E + \mathrm{div}(t) = B - F',$$

a rational replacement restricts to $-[E \cap F']$: B is disjoint from E, while F' meets it once. There is one pole and no zero, giving degree -1. A nonzero regular section of $\mathcal{O}_X(E)|_E$ could not have this negative degree.